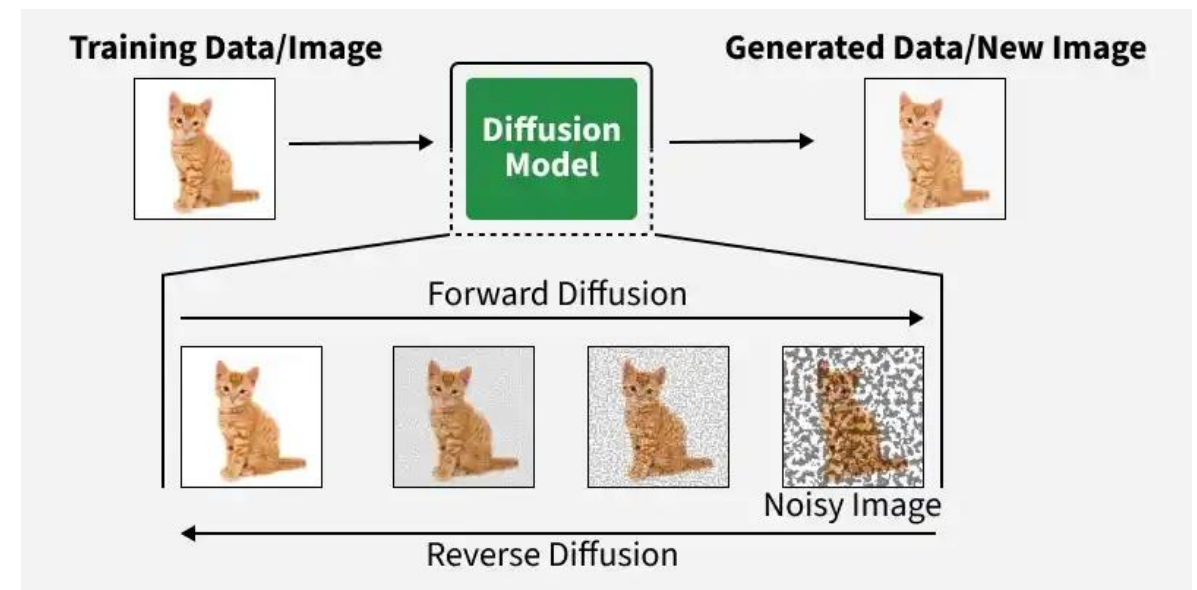
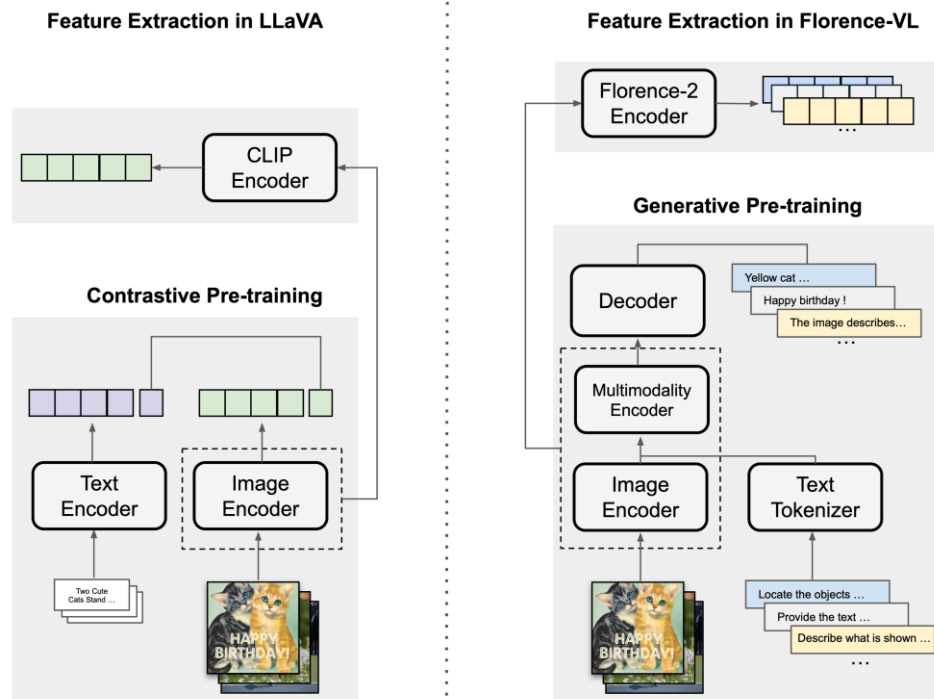


Diffusion Models: Generating Reality

- **Diffusion Process:** These models work by adding Gaussian noise to an image (forward) and learning to reverse the process to generate a clear image from scratch.
- **Stable Diffusion (SD 3.5 / SDXL 2026):** In 2026, these are used for **Medical Report Augmentation** (visualizing text descriptions) and **Synthetic Agriculture Data** (creating training images for rare crop diseases in KPK).
- **Visualization:** Essential for "What If" scenarios—visualizing how a crop will look if a specific pest spreads or how a tumor might respond to treatment.



Multimodal Giants: LLaVA, Florence-2, and Beyond



- **LLaVA (Large Language-and-Vision Assistant):** Connects a vision encoder (like CLIP successors) with a powerful LLM to enable "conversational sight."
- **Florence-2:** Microsoft's sequence-to-sequence foundation model that excels at high-precision tasks like segmentation and OCR in one unified framework.
- **Visual Agents:** Models like **Qwen2-VL** can now act as agents, "seeing" a computer screen and clicking buttons to complete tasks autonomously.

Beyond Accuracy: RAGAS & LLM-as-Judge

- **RAGAS**: The 2026 gold standard for RAG evaluation. It measures **Faithfulness** (is the answer grounded in context?), **Answer Relevancy**, and **Context Precision**.
- **LLM-as-Judge**: Using a larger "Reasoning Model" (like GPT-5 or DeepSeek-V3) to grade the performance of smaller models.
- **Faithfulness Metric**:
$$\text{Faithfulness} = \frac{|\text{Claims in Answer supported by Context}|}{|\text{Total Claims in Answer}|}$$

